

Ultimate Basic — C64 compiler

A modern BASIC-like language that compiles directly to 6502 machine code for the Commodore 64. Output: `.prg` files (VICE or real hardware) and `.d64` disk images.

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Quick start

```
cargo build --release

# Compile to .prg (memory map printed on success)
ub build demo.ub -o demo.prg

# Verbose: also prints zero-page layout and hex dump
ub build demo.ub -v

# Compile + create .d64 disk image
ub build demo.ub --d64 disk.d64

# Compile + create .d64 with extra files embedded
ub build demo.ub --d64 disk.d64 --add music.prg --add loader.prg
```

Language reference

Variables and constants

```
var x = 10           # 8-bit integer (default)
var ptr: word = $0400 # 16-bit – two zero-page bytes, usable as 16-bit address
var f: float = 3.5    # Q8.8 fixed-point – hi byte = integer, lo byte = fraction
var msg = "HELLO"     # string variable (pointer to inline PETSCII data)
var s: string = "TEXT" # string with explicit type
var scores = array(10) # byte array, 10 elements stored at $C000+
var times = array_word(8) # word array, 8 word elements stored at $C000+
const BORDER_ADDR = $D020 # compile-time constant (substituted inline, no ZP slot)
```

Keywords and identifiers are **case-insensitive**: `PRINT`, `Print`, and `print` are all valid.

Type	Width	Notes
<code>int</code>	8-bit	default for numeric literals
<code>word</code>	16-bit	two ZP bytes; can be used as address in <code>poke</code> / <code>peek</code>
<code>float</code>	16-bit Q8.8	hi byte = integer part (0–255), lo byte = fractional part
<code>string</code>	pointer	ZP pair → null-terminated PETSCII in code segment
<code>array(N)</code>	N bytes	byte elements; lives at <code>\$C000+</code> , not in ZP
<code>array_word(N)</code>	N×2 bytes	word (16-bit) elements; lives at <code>\$C000+</code> , not in ZP

Comments

```
# hash comment
rem this is also a comment
var x = 5 # inline comment
var x = 5 : var y = 6 # colon separates statements on one line
```

Operators

```
x = x + 1
y = a * b - c / 2
z = x and 15      # bitwise AND
w = a or b        # bitwise OR
v = a xor b       # bitwise XOR
m = x shl 3       # shift left
n = x shr 2       # shift right
r = x mod 40      # 8-bit modulo (remainder); SEC/SBC/BCS loop
b = bnot x        # bitwise NOT: x XOR 255 (complement all 8 bits)
```

Comparisons: `==` `!=` `<` `>` `<=` `>=` (return 1/0)

Increment / Decrement

```
inc x      # x = x + 1 (INC zp – single instruction)
dec x      # x = x - 1 (DEC zp – single instruction)
```

For `word` variables carry is handled: `inc` uses `INC lo; BNE skip; INC hi`; `dec` uses `LDA lo; BNE skip; DEC hi; DEC lo`.

Compound Assignments

```
x += 5          # x = x + 5
x -= 3          # x = x - 3
x *= 2          # x = x * 2
x /= 4          # x = x / 4
x and= 15       # x = x and 15  (bitwise AND)
x or= 64        # x = x or 64   (bitwise OR)
x xor= 255      # x = x xor 255 (bitwise XOR)
x shl= 2        # x = x shl 2
x shr= 1        # x = x shr 1
```

Print

```
print "HELLO"
print x
print x, y, "text"
print          # blank line

print spc(5)    # print 5 space characters
print tab(20), "VALUE" # move cursor to column 20, then print
print "A", spc(3), "B" # mix freely

print "A=" + a    # string + numeric var
print s1 + s2     # two string vars → sequential print
print "Hello " + "World" # two literals → compile-time fold
print chr$(13)    # print by PETSCII code
```

chr\$

```
print chr$(65)    # output character with PETSCII code 65 ('A')
var c = chr$(n)   # store byte value n in variable c
print ">" + chr$(42) # usable in string concat
```

Branching

```

if x == 1 then
    print "YES"
else
    print "NO"
end

```

Select / Case

```

select x
case 1:
    print "ONE"
case 2:
    print "TWO"
else:
    print "OTHER"
end

```

`select expr` evaluates the expression once and compares it against each `case` value in order. The first matching case body is executed and control jumps to after `end`. The optional `else:` body runs if no case matches. All values must be 8-bit (0–255).

Loops

```

loop 5                # counted loop (5 iterations)
    print "HI"
end

times 5               # alias for loop N ... end
    print "HI"
end

loop                  # infinite loop
    x = x + 1
    if x == 5 then continue end # skip to next iteration
    if x == 100 then break end
end

for i = 1 to 10        # for..next (preferred)
    if i == 5 then continue end # skip to increment step
    print i
next

```

```

for i = 0 to 20 step 2
  print i
next i          # variable name after 'next' is optional

loop i = 1 to 10    # legacy loop..end syntax – identical code
  print i
end

while x < 100
  x = x + 1
end

repeat          # do-while: body runs at least once
  x = x + 1
until x == 100   # exits when condition is true

```

Labels and goto

```

label main_loop
  x = x + 1
  if x < 10 then goto main_loop end

```

Forward `goto` (label defined later) is fully supported.

`gosub label` / `return` jump to a label and return back (JSR / RTS at machine code level). The label must be a `label name` statement, not a `sub` — it has no parameters and shares the same zero-page scope. `gosub` supports forward references (label defined after `gosub`).

```

gosub draw_border
...
label draw_border
  # ... draw something ...
  return          # RTS – returns to the instruction after gosub

```

Subroutines

```

sub greet()
  print "HELLO!"

```

```

end

sub set_color(col)
    color border col
    color text   col
end

greet()          # call with parens
call greet       # call keyword (no parens)
set_color(6)

```

Parameters are passed via dedicated zero-page slots. No recursion (slots are static).

Typed parameters are supported: `sub draw(x, y:int)` or `sub copy(src:string)` — string parameters receive a 2-byte pointer so the callee can index the source string via `src[i]`.

Functions (return values)

```

fn square(x)
    return x * x
end

fn add(a, b)
    return a + b
end

fn clamp(val, lo, hi)
    if val < lo then return lo end
    if val > hi then return hi end
    return val
end

var s = square(9)          # s = 81  - fn call as expression
var c = clamp(joy, 0, 39)  # c = clamped value
print add(10, 20)         # 30  - usable inline in print

```

Functions support an optional `: word` return type for 16-bit values:

```

fn get_addr(): word
    return $C000
end

var ptr: word = get_addr() # ptr = $C000
poke ptr, 42              # STA (ptr),Y - valid indirect addressing
var v = peek(ptr)         # LDA (ptr),Y

```

`fn` is emitted in pass 2 (same as `sub`), so function bodies are never executed at startup. Forward references are fully supported.

Arrays

```
var scores = array(8)    # 8 bytes at $C000

scores[0] = 100          # constant index → STA $C000
scores[i] = 99           # variable index → STA (ptr),Y
var v = scores[i]        # LDA (ptr),Y
print scores[2]          # usable inline in print

var times = array_word(8) # 16 bytes (8×2) at $C000+

times[0] = $1234         # constant index → STA $C000 (lo), STA $C001 (hi)
times[i] = $5678         # variable index → ASL A for stride; (ptr),Y × 2
var t: word = times[1]   # LDA $C002, LDA $C003
```

16-bit (word) variables

```
var ptr: word = $0400    # two ZP bytes: lo=$00 hi=$04
poke ptr, 6              # STA (ptr),Y
var v = peek(ptr)        # LDA (ptr),Y
```

Bitmap graphics

```
graphics on              # VIC-II hires bitmap mode (320×200, 1bpp); bitmap at $2000
graphics on multi        # VIC-II multicolor bitmap mode (160×200, 2bpp, 4 colours/cell)
graphics off             # return to text mode

gcls                     # clear bitmap (fills $2000-$3FFF) + set video matrix colors

plot x, y                # set pixel at (x, y); x: 0-319, y: 0-199
plot erase x, y           # clear pixel (AND ~mask)
plot xor x, y            # toggle pixel (EOR mask) – flicker-free animation
circle x, y, r           # midpoint circle centered at (x, y) with radius r; clips off-screen points
line x1, y1, x2, y2      # Bresenham line from (x1,y1) to (x2,y2); x: 0-255, y: 0-199
paint x, y               # 4-connected flood fill from (x, y); fills clear pixels bounded by set ones
```

```

mplot x, y, color      # set multicolor pixel (x: 0-159, y: 0-199, color: 0-3); requires graphics on
multi

```

Both `graphics on` variants blank the display (`LDA $D011 / AND #$EF / STA $D011`) while switching VIC registers, then re-enable it in the target mode — prevents mode-switch glitches.

Block graphics (80×50)

```

graphics on block      # 80×50 block-pixel mode (text mode + custom 4-pixel charset @ $2800)
graphics off           # return to text mode
gcls                   # clear block playfield: screen RAM $0400-$07FF + color RAM $D800-$DBFF

plot4 x, y             # set block pixel at (x, y); x: 0-79, y: 0-49
plot4 erase x, y        # clear block pixel at (x, y)
circle4 x, y, r        # draw midpoint circle in block pixels; clips to 80×50

```

A chunky low-res mode layered on standard 40×25 text. A 16-character custom charset copied to `$2800` encodes a 2×2 quadrant grid per character (bit3=TL, bit2=TR, bit1=BL, bit0=BR), so each text cell holds 2×2 block pixels → an effective 80×50 grid. No bitmap RAM is used (`$2000-$3FFF` stays free), making it faster than hires bitmap. `plot4` OR's the quadrant bit into the cell so overlapping pixels accumulate; `plot4 erase` clears it. `circle4` uses the same block-pixel helper to draw an outline circle in the 80×50 coordinate space. `gcls` clears both screen and color RAM. See `examples/block_demo.ub`.

Screen and color

```

cls                   # clear screen (KERNAL $E544)
cls fast              # fast fill: screen RAM + color RAM + HOME

color text 14         # text color register $0286
color border 6        # $D020
color bg 0            # $D021

screen 0, 0, 65       # write char 65 ('A') to screen RAM at col 0, row 0 ($0400)
screen 10, 5, ch      # col 10, row 5 – col/row can be variables
screen 5, 3, 42, 7    # char 42 at col 5, row 3, color 7 (writes color RAM $D800 too)
screen x, y, ch, col  # all four arguments as variables

display on           # re-enable VIC display ($D011 DEN bit)
display off          # blank display

lowercase             # CHR$(14) → switch VIC-II to lowercase/uppercase charset
uppercase             # CHR$(142) → switch VIC-II back to uppercase/graphics charset

```

```

cursor 20, 10      # move cursor to col 20, row 10 (KERNAL PLOT $FFF0)
cursor x, y        # column from variable x (0-39), row from y (0-24)

print at 20, 10, "HELLO" # cursor(20,10) + print in one statement (no trailing newline)
print at x, y, "Score:", score # any mix of exprs
print at 0, 0      # position only (no text, no newline)

scroll x 3          # set horizontal fine scroll: $D016 bits 0-2 = 3 (0-7)
scroll y 2          # set vertical fine scroll: $D011 bits 0-2 = 2 (0-7)
scroll x n          # value can be a variable or expression (masked to bits 0-2)
scroll x 7 narrow   # set fine scroll and force 38-column mode (hide edge column)
scroll x 0 wide     # set fine scroll and restore 40-column mode
scroll row 12 left  # shift screen RAM row 12 left by one character

```

`lowercase` emits `LDA #$0E; JSR $FFD2` at runtime. String literals compiled after `lowercase` have their case swapped automatically — uppercase source chars are stored in the PETSCII lowercase slot (`$61+`) and lowercase source chars in the uppercase slot (`$41+`) — so `"Hello World"` source displays as **Hello World** on screen. `uppercase` emits `LDA #$8E; JSR $FFD2` and reverts to direct mapping. `cls` does **not** reset the charset mode.

`scroll x n` writes `(n AND 7)` into bits 0-2 of `$D016` (preserving bits 3-7).

`scroll x n narrow` writes the fine-scroll bits and clears `$D016` bit 3 (38-column mode).

`scroll x n wide` writes the fine-scroll bits and sets `$D016` bit 3 (40-column mode).

`scroll y n` writes `(n AND 7)` into bits 0-2 of `$D011` (preserving bits 3-7).

`scroll row R left` shifts one constant screen row left; write the new rightmost character with `screen 39, R, ch`.

Useful for smooth hardware scrolling: decrement each frame from 7 down to 0, shift screen RAM, reset to 7.

`screen col, row, char [, color]` writes directly to screen RAM (`$0400 + row*40 + col`) and optionally to color RAM (`$D800 + row*40 + col`). Constant col/row: address computed at compile time.

Ultimate 64 — CPU Speed

```

speed 4            # set CPU to 4 MHz (reads $D031, updates bits 0-3, writes back)
speed 48           # 48 MHz (maximum speed on U64)
speed max          # same as speed 48 (alias)
speed off          # back to 1 MHz (alias for speed 1)

badlines on        # enable badline timing ($D031 bit 7 = 0, default C64 behaviour)
badlines off       # disable badline timing ($D031 bit 7 = 1, more CPU cycles)

var t = turbo()    # 1 if turbo is active (bits 0-3 of $D031 != 0), 0 if at 1 MHz

```

Available MHz values: 1, 2, 3, 4, 5, 6, 8, 10, 12, 14, 16, 20, 24, 32, 40, 48.

Constant values are rounded down to the nearest available speed at compile time.

Variable values are treated as a raw speed index (0-15) and OR'd into bits 0-3 of `$D031`.

\$D031 index	MHz (U64)	MHz (U64 Elite-II)
0	1	1
3	4	4
6	8	10
11	20	24
15	48	64

Requires **U64 Turbo Control** set to **U64 Turbo Registers** or **Turbo Enable Bit** in the U64 config menu.
On a regular C64 or emulator without the register, the **poke** to **\$D031** is silently ignored.

Keyboard

```

var key = getch()      # busy-wait on $FFE4 until key; returns PETSCII code
var k   = inkey()      # non-blocking: returns PETSCII code, or 0 if no key pressed
waitkey                # wait until any key is pressed (CIA1 matrix scan; works during IRQ)
var k   = waitkey()    # same but returns raw $DC01 column bits (0 bit = key pressed)
var j = joy(2)         # read joystick port 2; returns inverted bits 0-4
var j = joy(1)         # read joystick port 1
                        # bit0=up(1) bit1=down(2) bit2=left(4) bit3=right(8) bit4=fire(16)

var mx = mouse_x()     # 1351 mouse X: SID POT X ($D419), 0-255
var my = mouse_y()     # 1351 mouse Y: SID POT Y ($D41A), 0-255
var mb = mouse_btn()   # mouse buttons: bit0=left (fire), bit1=right

```

Exit

```

bye                # JSR $E544 (clear screen), clear STOP flag, RTS to BASIC
exit               # alias for bye

```

Timing

```

wait 50            # wait 50 raster-line transitions (~3.2 ms)
wait raster 100    # spin until $D012 == 100 (raster-split effects)
delay 1            # wait 1 PAL frame (1/50 s ≈ 20 ms)
delay 20           # wait 20 frames ≈ 0.4 s; n can be a variable (0-255)

```

delay N counts N complete PAL frames using raster line 200 as the frame boundary.

SID Sound

```
sound 0, $1CAD, 25      # voice 0, freq $1CAD (~ middle C PAL), 25 frames duration
sound 1, freq_word, 50   # voice 1, freq from word var, 50 frames (1 s at 50 Hz)
sound 2, 0, 0           # voice 2, silence

sid volume 15           # master volume full ($D418 = $0F); range 0-15
sid volume 0            # silence (master volume = 0)
sid stop                # zero all 25 SID registers ($D400-$D418) – complete silence
```

sound <channel>, <freq>, <duration> — duration in PAL frames (1/50 s each).

Fixed ADSR: attack/decay \$09, sustain/release \$F0, sawtooth waveform.

Master volume \$D418 always set to \$0F.

sid volume N writes N to \$D418. Bits 0-3 = volume (0-15), bits 4-7 = filter mode.

sid stop emits a 10-byte zero-fill loop — faster than 25 individual pokes.

Music playback

music play/stop/pause/resume is a high-level alternative to the manual sys sid_init / cia_timer setup.

Requires a prior load sid statement (defines sid_init / sid_play).

```
load sid "tune.sid"      # embed SID file (defines sid_init / sid_play)

music play               # initialise sub-tune 0 + start CIA1 50 Hz IRQ
music play 1             # start from sub-tune 1 (song number 0-based)
music stop               # stop playback + zero all 25 SID registers ($D400-$D418)
music pause              # disable CIA1 timer A IRQ (music freezes, SID unchanged)
music resume             # re-enable CIA1 timer A IRQ (continues from pause point)
```

Statement	Effect
music play [n]	call sid_init(n), configure CIA1 timer A at 19 656 cycles (~50 Hz PAL), install IRQ wrapper
music stop	disable CIA1 IRQ + zero all 25 SID registers
music pause	disable CIA1 IRQ (SID output remains frozen)
music resume	re-enable CIA1 IRQ (continues from pause point)

The IRQ wrapper (emitted once at the end of the program) does: ACK CIA1 timer A → JSR sid_play → JMP \$EA81.

Sprites

```

sprite 0, x, y, $2000    # sprite 0: set X, Y position and data pointer
sprite 0, x, y           # without data pointer (keeps existing)
sprite on 0              # enable sprite 0 ($D015 |= bit0)
sprite off 0             # disable sprite 0 ($D015 &= ~bit0)
sprite color 0, 7        # sprite 0 color = yellow ($D027)
sprite multicolor 0, on  # enable multicolor mode for sprite 0 ($D01C |= bit0)
sprite multicolor 0, off # disable multicolor mode ($D01C &= ~bit0)
sprite expand x 0, on    # double width ($D01D |= bit0)
sprite expand x 0, off   # normal width ($D01D &= ~bit0)
sprite expand y 0, on    # double height ($D017 |= bit0)
sprite expand y 0, off   # normal height ($D017 &= ~bit0)
sprite priority 0, on    # behind background ($D01B |= bit0)
sprite priority 0, off   # in front of background ($D01B &= ~bit0)
var h = sprite_hit()     # sprite-sprite collision ($D01E, cleared on read)
var b = sprite_bg_hit()  # sprite-background collision ($D01F, cleared on read)
var x = sprite_x(0)      # read sprite 0 X position (lo byte, $D000)
var y = sprite_y(0)      # read sprite 0 Y position ($D001)
sprite_frame 0, $2000    # set sprite 0 data pointer only: $07F8+id = $2000>>6
                        # (does NOT change X/Y position)

```

X supports full 9-bit range (0–319): use a `word` variable for runtime values > 255.

Sprite data pointer: `data_addr` must be 64-byte aligned; stored as `addr >> 6` at `$07F8+id`.

Sprite definition

```

sprdef 0
    $00,$3C,$00,  $00,$FF,$00,  $03,$FF,$C0,  $07,$FF,$E0,
    $0F,$FF,$F0,  $0F,$FF,$F0,  $1F,$FF,$F8,  $1F,$FF,$F8,
    $1F,$FF,$F8,  $0F,$FF,$F0,  $0F,$FF,$F0,  $07,$FF,$E0,
    $03,$FF,$C0,  $00,$FF,$00,  $00,$3C,$00,  $00,$00,$00,
    $00,$00,$00,  $00,$00,$00,  $00,$00,$00,  $00,$00,$00,
    $00,$00,$00
end

```

`sprdef id ... end` embeds 63 sprite bytes at the next 64-byte-aligned address in the code segment, emits a `JMP` over them, and automatically sets `$07F8+id = data_addr >> 6`.

To use the same shape for multiple sprites, read back the pointer:

```

var pg = peek($07F8)    # pointer set by sprdef 0
poke $07F9, pg          # copy to sprites 1–7

```

Custom charset

```
charset $3800          # set base address for chardef (default $3800)

chardef 65             # redefine character 65 ('A')
    $18,$3C,$66,$7E,$66,$66,$66,$00
end

chardef 66             # fewer than 8 bytes are zero-padded
    $7C,$66,$7C,$66,$7C
end
```

`charset base` sets the destination address used by all subsequent `chardef` statements. `chardef id ... end` embeds 8 bytes inline in the code segment (preceded by a `JMP` to skip over them), then copies them to `charset_base + id*8` at runtime. Values must be compile-time constants; use `%` for binary literals (`%00011000`).

To activate a custom charset in VIC-II, set the character generator address via `$D018` :

```
charset $3800
chardef 1 $FF,$81,$81,$81,$81,$81,$81,$FF end # box border
poke $D018, $1A    # screen at $0400, charset at $3800 (bank 0)
```

Memory

```
poke $D020, 2        # STA $D020
poke addr_var, 6      # STA (addr_var),Y - if addr_var is word type
var v = peek($D012)   # LDA $D012
var v = peek(addr_var) # LDA (addr_var),Y - if addr_var is word type

var w: word = peek16($C000) # read 16-bit little-endian: lo=$C000, hi=$C001
poke16 $0314, $EA81      # write 16-bit little-endian: lo-$0314, hi-$0315
poke16 ptr, w            # word var as address; word var as value
```

`peek16(addr)` reads two consecutive bytes (lo, hi) as a `word` . `poke16` writes lo then hi.

Disk I/O

```

load "PROGRAM"          # KERNAL LOAD: loads file from device 8 to its native address
load "DATA", $C000       # loads file to a specific address
load "DATA", ptr         # addr from word variable

save "DATA", $C000, 4096 # KERNAL SAVE from $C000, 4096 bytes → device 8
save "PROG", start, len  # addr and len from word/int variables

```

`load` calls KERNAL `SETNAM + SETLFS + LOAD ( $FFBD / $FFBA / $FFD5 )`.

Without address: secondary address 0 (file's own 2-byte header used as load address).

With address: secondary address 1 (file loaded to specified location).

`save` calls `SETNAM + SETLFS + SAVE ( $FFBD / $FFBA / $FFD8 )`. Requires both `addr` and `len`.

SID Music

```

load sid "tune.sid"      # embed SID music at its native load address
load sid "tune.sid", $2000 # override: embed at $2000 regardless of SID header

```

`load sid` reads a PSID or RSID file at **compile time**, strips the header, and appends the raw music bytes to the output `.prg`. After `load sid`, two compile-time constants become available:

Constant	Description
<code>sid_init</code>	Init routine address — call once with A = song number (0-based)
<code>sid_play</code>	Play routine address — call every frame (50 Hz PAL) from an IRQ handler

Both constants work anywhere a constant address is accepted: `sys`, `irq`, `poke`, expressions.

Typical usage:

```

load sid "music.sid"

sub music_irq()
    poke $D019, $FF    # ACK VIC raster IRQ
    sys sid_play        # advance one frame of music
    irq_exit            # JMP $EA81: restore A/X/Y + RTI (proper IRQ exit)
end

sys sid_init, 0        # initialise SID chip: A=0 → first sub-tune
irq music_irq, $C0     # raster IRQ at line $C0 → 50 Hz on PAL

sid volume 15          # master volume on

```

Notes:

- SID data is placed **after** all generated code, padded with zeros up to the load address. The compiler warns if the SID load address would overlap generated code.
- PSID v1 and v2 are supported. If the SID header's load address is 0, the first two data bytes are used as the address (PRG-style, little-endian).
- Only one `load sid` per program is meaningful (the last one wins).

Serial channel file I/O

```
open 1, 8, 2, "MYFILE" # open logical file 1, device 8, secondary 2, name "MYFILE"
open 2, 4, 7           # open printer (device 4), no filename
open ch, dev, sec      # channel, device, secondary from variables

print# 1, "HELLO"      # send "HELLO"+CR to logical file 1
print# ch, x, "text"    # any mix of vars, strings – same as print but to file

close 1                # close logical file 1
close ch               # channel from variable
```

`open` calls `SETNAM` (\$FFBD) + `SETLFS` (\$FFBA) + `OPEN` (\$FFC0). Without a filename, `SETNAM` is called with length 0. `print#` routes output via `CHKOUT` (\$FFC9), `CHROUT` per char (+ trailing CR), then `CLRCHN` (\$FFCC). `close` puts the channel number in A and calls `CLOSE` (\$FFC3).

Input

```
input score            # read up to 3 digits from keyboard → 8-bit int var
input "Name: ", name   # optional prompt string, then read line → string var
input "Score: ", score  # prompt + int input
```

`input` uses `KERNAL BASIN` (\$FFCF) for blocking, echoed line input with DEL support.

- **Int var:** accepts only 0–9, max 3 chars; converts to 8-bit value on CR.
- **String var:** accepts up to 30 chars; stores as null-terminated string; ZP pair updated.

Float / Fixed-Point

`float` variables use Q8.8 fixed-point format: the high byte is the integer part (0–255) and the low byte is the fractional part (0/256 ... 255/256).

```

var f: float = 3.5      # 3.5 → hi=3, lo=128 (= 0x0380)
var g: float = 0        # integer 0 is promoted to 0.0 automatically

f = 1.5                # Q8.8 literal assignment
f = f + 1.5            # 16-bit Q8.8 arithmetic (result: 3.0)
f = f + g              # float + float

var n = int(f)          # extract integer part (hi byte) → 8-bit int
print f                # prints as "N.DD" (e.g. 3.5 → "3.50", 1.25 → "1.25")

```

Operation	Example	Notes
Literal	3.5, 0.25, 1.0	parsed as Q8.8 at compile time
Integer promotion	f = 5	stores 5.0 (hi=5, lo=0)
Add/sub	f + 1.5, f - g	16-bit Q8.8 arithmetic
Extract int	int(f)	returns hi byte as 8-bit int
Print	print f	format "N.DD", always 2 fractional digits

Caveat: Arithmetic overflow wraps at 255.255 (no saturation). Multiplication and division of two float variables are not yet supported — use `int()` + integer arithmetic for those cases.

Math functions

```

var a = abs(x - 20)     # two's-complement absolute value
var b = min(x, 39)      # 8-bit minimum
var c = max(x, 0)       # 8-bit maximum
var s = sgn(score)      # 0 = zero, 1 = positive (1-127), $FF = negative (128-255)
var r = rnd()           # LCG pseudo-random 0-255; seed from raster line
var r = rnd(10)         # LCG pseudo-random 0-9 (rnd() mod n; result 0..n-1)
var s = sin(angle)      # sine: angle 0-255 (full circle), returns 0-255 (center=128)
var c = cos(angle)      # cosine = sin(angle+64)
var v = clamp(x, 0, 39) # 8-bit unsigned clamp: result = max(lo, min(hi, val))

print hex(n)           # print as 2-digit uppercase hex
print bin(n)           # print as 8-bit binary string

```

String functions

```

var n = len(msg)      # length of null-terminated string var (0-255)
var c = asc(msg)      # PETSCII code of first character (0 if empty)
var c = asc("A")      # compile-time: constant PETSCII code
var n = val(s)        # runtime: parse decimal PETSCII string → 8-bit int (e.g. "042" → 42)
var c = msg[i]        # string character at index i: PETSCII code of msg[i]
msg[i] = c            # write PETSCII byte c to string at index i – STA (ptr),Y
msg[0] = 72           # constant index → LDY #0; STA (ptr),Y

```

Number formatting

```

print hex(n)          # print as 2-digit uppercase hex
print bin(n)          # print as 8-bit binary string
print dec(n, 4)       # right-justified decimal in a field of 4 chars (e.g. 42 → " 42")
print dec(n, width)   # width can also be a variable

```

`dec(n, width)` pads the number on the left with spaces to fill `width` characters.

If the number has more digits than `width`, it is printed without padding (no truncation).

In non-print contexts `dec(n, w)` evaluates to `n` unchanged (same as `hex` / `bin`).

REU (RAM Expansion Unit)

```

var ok = reu_present() # 1 if REU detected, 0 if not (write/read test on $DF04)
var ok = reudet()      # alias for reu_present()

reu stash c64addr, bank, reu_addr, len # copy C64 → REU
reu fetch c64addr, bank, reu_addr, len # copy REU → C64
reu swap c64addr, bank, reu_addr, len # swap between C64 and REU

```

`reu_present()` performs a write/read-back test on REU register `$DF04`. Without an REU the write is lost (open bus), so the read-back differs — reliably detects presence without touching any side-effecting command register.

Parameter	Width	Notes
<code>c64addr</code>	16-bit	C64 RAM start — constant, <code>word</code> var, or 8-bit expr
<code>bank</code>	8-bit	REU bank number (0–7 for a 512 KB unit)
<code>reu_addr</code>	16-bit	Offset within the REU bank

Parameter	Width	Notes
len	16-bit	Bytes to transfer (0 = 65 536 in REU hardware)

REU registers: \$DF01 command (\$B0 stash / \$B1 fetch / \$B2 swap),
\$DF02–\$DF03 C64 addr, \$DF04–\$DF05 REU offset, \$DF06 bank, \$DF07–\$DF08 length.
Transfer is synchronous (CPU halted during DMA).
Requires a real REU or VICE: **Settings** → **Hardware** → **RAM Expansion Module**.

Memory utilities

```
fill screen 32          # fill screen RAM $0400-$07FF with value 32 (space char)
fill color 1            # fill color RAM $D800-$DBFF with value 1 (white)

fill $0400, 1000, 32    # fill 1000 bytes starting at $0400 with value 32
fill addr, 256, 0       # addr can be word var
fill ptr, len_word, val # all operands can be expressions / word vars

memcpy $C000, $0400, 256 # copy 256 bytes from $C000 → $0400
memcpy src_ptr, dst_ptr, 40 # word vars for source and destination

drawmem $C000, $0400, 8, 10, 40 # blit 8x10 rect from $C000 → screen at $0400, stride 40
drawmem src_ptr, dst_ptr, w, h, 40 # word vars for src/dst
```

Both `fill` and `memcpy` support 16-bit lengths (0–65535). Use `word` variables for lengths > 255.

`drawmem src, dst, width, height, stride` copies a 2-D rectangular block. `src` is read linearly (packed rows); `dst` advances by `stride` bytes between rows — use `40` (\$28) for the C64 screen or color RAM (40 columns). Width, height and stride are all 8-bit values.
`src` and `dst` may be constants, `word` variables, or 8-bit expressions.

Raster IRQ

```
irq my_handler          # raster IRQ at line 0, handler = sub name or address
irq my_handler, 100     # raster IRQ at raster line 100
irq $C800, 200          # handler at fixed address
irq addr_word           # handler address from a word variable
```

Sets up a raster IRQ via the BASIC soft vector (\$0314 / \$0315): disables CIA1 timer IRQ, ACKs pending VIC IRQ, writes raster line to \$D012, enables VIC raster IRQ (\$D01A=\$01), writes handler address, and re-enables interrupts.

The handler **must** end with `sys $EA81` (KERNAL end-of-IRQ) — plain `RTS` or `RTI` will corrupt the stack. ACK the VIC IRQ first:

```

sub my_handler()
    poke $D019, $FF      # ACK VIC IRQ
    # ... work here ...
    sys $EA81            # JMP to KERNAL end-of-IRQ
end

```

Forward references are supported (`irq my_handler` before the sub is defined).

NMI handler

```

nmi my_nmi                # set NMI vector $0318/$0319 to handler sub or address

sub my_nmi()
    # ... NMI work here ...
    nmi_exit                # JMP $FE47 — proper NMI exit (restores A/X/Y + RTI)
end

```

`nmi handler` writes the handler address to the NMI soft vector (`$0318 / $0319`). The hardware NMI vector at `$FFFA` points to the KERNAL NMI routine which branches through `$0318` . The handler **must** end with `nmi_exit` (emits `JMP $FE47`) — using plain `RTI` will corrupt the stack. Forward references supported.

CIA1 timer IRQ

```

cia_timer 19656, my_handler # CIA1 timer A: fires every 19656 cycles (~50 Hz PAL)
cia_timer period, handler   # period can be a variable or expression

```

Sets up CIA1 timer A as a periodic IRQ source via the BASIC soft vector (`$0314 / $0315`):

1. SEI — disable interrupts
2. `$DC0D = $7F` — disable all CIA1 IRQs
3. Load 16-bit period lo→ `$DC04` , hi→ `$DC05`
4. Write handler address to `$0314 / $0315`
5. `$DC0D = $81` — enable CIA1 timer A IRQ
6. `$DC0E = $01` — start timer A in continuous mode
7. CLI — re-enable interrupts

The handler must end with `irq_exit` (or `sys $EA81`) and should ACK the CIA1 IRQ:

```

sub my_handler()
  poke $DC0D, $01      # ACK CIA1 timer A IRQ (read also clears it)
  # ... work here ...
  irq_exit             # JMP $EA81: restore A/X/Y + RTI
end

```

PAL timing: clock = 985 248 Hz. Period for 50 Hz = 985 248 / 50 = 19 705 cycles ≈ \$4CC9 . Forward references supported.

Error handling

```

onerr goto err_handler  # set KERNAL I/O error vector ($0300/$0301) to a label

...

label err_handler
  print "I/O ERROR"
  bye

```

`onerr goto label` writes the label address (lo, hi) to KERNAL locations \$0300 and \$0301 .

When a KERNAL I/O error occurs (e.g. a failed `load` or `open`) the KERNAL executes `JMP ($0300)` , which branches to the label. Forward references (label defined after `onerr goto`) are supported.

Compile-time file embedding

```

incbin "sprites.bin"    # embed raw binary bytes at current code position
include "defs.ub"       # inline another .ub source file (lexed+parsed in place)

```

Data / Read

```

data 1, 2, 3, 255      # constant byte table
read varname           # load next byte into varname (auto-declares if needed)

```

All `data` values are collected at compile time. A 2-byte ZP pointer is automatically allocated and initialised at program start. Each `read` advances the pointer.

Inline assembly

```
sys $FFD2          # JSR $FFD2
sys $FFD2, 7       # LDA #7 ; JSR $FFD2 (pass byte value in A register)
irq_exit           # JMP $EA81 – proper IRQ handler exit (restore A/X/Y + RTI)
asm $EA, $EA       # inline raw bytes (NOP NOP) – legacy form
asm {
    ; Full 6502 mnemonics and addressing modes
    LDA #$07        ; immediate
    STA $0286       ; absolute
    LDA $50         ; zero-page ($50 ≤ $FF → ZP auto-selected)
    STA $0400,X     ; absolute,X indexed
    LDA ($50),Y     ; (indirect),Y
    LDA ($50,X)     ; (indirect,X)
    JSR $FFD2       ; subroutine call
    JMP $C000       ; absolute jump
    JMP ($FFFC)     ; indirect jump

    CLC
    ADC #1
    SEC
    SBC #1

    TAX             ; implied / transfer
    ASL A           ; accumulator (also just: ASL)
    LSR A
    ROL
    ROR

    ; Branches – operand is an absolute address; offset is computed automatically
    BNE loop        ; forward or backward branch to local label
    BEQ done

loop:
    NOP
done:
    RTS

    ; #<label / #>label – lo / hi byte of a label address
    LDA #<handler
    STA $0314
    LDA #>handler
    STA $0315

    ; * – current assembly location
    JMP *

    ; Raw hex bytes (backward-compatible with old asm { $xx ... } syntax)
    $EA $EA        ; two NOP bytes
}
```

Addressing modes:

Syntax	Mode	Bytes	Example
(no operand)	Implied	1	NOP, RTS
A	Accumulator	1	ASL A, LSR
#value	Immediate	2	LDA #\$07
\$zz (0–255)	Zero-page	2	LDA \$50
\$zz,X	ZP,X	2	LDA \$50,X
\$zz,Y	ZP,Y	2	LDX \$50,Y
\$xxxx	Absolute	3	LDA \$0400
\$xxxx,X	Absolute,X	3	LDA \$0400,X
\$xxxx,Y	Absolute,Y	3	LDA \$0400,Y
(\$xxxx)	Indirect	3	JMP (\$FFFC)
(\$zz,X)	(Indirect,X)	2	LDA (\$50,X)
(\$zz),Y	(Indirect),Y	2	LDA (\$50),Y
label	Relative	2	BNE label (branches only)

- `$zz` (1–2 hex digits, value ≤ 255) selects zero-page if the instruction supports it; otherwise auto-upgrades to absolute. Use `$00xx` (4 digits) to force absolute.
- Branch operands are absolute addresses; the relative byte offset is computed automatically.
- Local labels (`name:`) are scoped to the `asm { }` block. Forward branches resolved in pass 2.
- `#<label` / `#>label` yield the lo / hi byte of a label's address.
- `*` yields the current instruction address, so `JMP *` assembles as a self-loop.
- Lines starting with `$`, `%`, or a digit are emitted as raw bytes (backward-compatible).
- Inside `asm { }`, comments are `;` or `//` to end of line. (`#` is the immediate prefix, not a comment.)

Mixing `asm { }` with subroutine parameters

Parameter names are **not accessible** inside `asm { }` blocks. Use `UltimateBasic` statements to move values into known locations before the `asm { }` block:

```
sub set_colors(border_col, bg_col)
  poke $D020, border_col  # UltimateBasic resolves the ZP address
  poke $D021, bg_col
  asm {
    ; values are already in $D020 / $D021
    LDA $D020
  }
end
```



For routines whose entire body is assembly — especially IRQ handlers that must cross-reference each other — put all handlers in a **single top-level `asm { } block`** in the main program. Labels in the same block share scope, so `irq1` and `irq2` can reference each other freely. See `examples/raster_irq_demo.ub`.

String ↔ integer

```
numstr score, $0340      # writes "042\0" at $0340 (always 3 digits, zero-padded)
var n = str_to_int("42") # compile-time: Expr::Number(42)

print str$(score)        # print 8-bit int as 3-digit decimal string ("000"-"255")
print "Score: " + str$(score) # usable in string concat print context
var s: string = str$(n)   # assign str$() result to a string var (shared static buffer)
```

`str$(n)` converts an 8-bit value to a 3-character decimal string (always 3 digits with leading zeros, e.g. `5` → `"005"`, `42` → `"042"`, `255` → `"255"`) followed by a null terminator. The result pointer is stored in a permanent ZP pair allocated at compile time.

Note: `str$(n)` uses a single shared 4-byte static buffer. Calling `str$(n)` again overwrites the previous result. For concurrent display of multiple values, use `numstr` to write to separate absolute addresses.

Examples

File	Description
<code>examples/features.ub</code>	const, label/goto, poke/peek, rnd, math functions
<code>examples/new_features.ub</code>	sub params, arrays, word vars, string vars
<code>examples/bitmap_demo.ub</code>	320×200 bitmap, plot, graphics on/off
<code>examples/block_demo.ub</code>	80×50 block graphics, plot4, circle4, graphics on block
<code>examples/joystick_demo.ub</code>	joystick reading, sprite movement
<code>examples/mux_demo.ub</code>	raster sprite multiplexer (3 windows × 8 sprites = 24)
<code>examples/orbit_demo.ub</code>	24-sprite orbit with pulsating radius and random colors
<code>examples/plasma_demo.ub</code>	plasma-effect bitmap with raster bar border animation
<code>examples/sprite_data.ub</code>	sprdef shape data (included by other demos)
<code>examples/sprite_mux_orbit.ub</code>	24-sprite orbit demo with sprdef + precomputed positions
<code>examples/sprite_orbit_demo.ub</code>	8 hardware sprites in circular orbit via sin/cos table

File	Description
examples/reu_bitmap_demo.ub	REU stash/fetch with bitmap graphics
examples/sid_music_demo.ub	SID music player with raster IRQ and keyboard exit
examples/tenprint.ub	5 TENPRINT maze implementations with menu; demos lowercase charset mode

Architecture

```
.ub source
  → Lexer → Vec<Token>
  → Parser → Vec<Stmt>
  → Codegen → Vec<u8> (raw 6502 machine code)
  → mod.rs → PRG = BASIC SYS stub + machine code
```

Two-pass compilation: Pass 1 = main code, Pass 2 = subroutine bodies (appended after main `RTS`). Forward references (`call` , `goto` , `plot`) are patched at the end.

Zero-page layout: `$02-$4F` permanent (vars, loop counters, sub params), `$50-$7F` scratch (reset per statement), `$FB` RNG seed.

CLI reference

```
ub build <input.ub> [OPTIONS]

-o, --output <file>  Output .prg file (default: <input>.prg)
-v, --verbose        Show zero-page layout and code hex dump
--no-stub            Skip the BASIC SYS stub (code loads at $0801)
--d64 [file]         Also produce a .d64 disk image;
                     without a filename defaults to <output>.d64
--add <file>         Add an extra file to the .d64 disk image;
                     may be repeated for multiple files
-h, --help           Show help
```

`--add` requires `--d64`. The compiled `.ub` program is always the first file on the disk; each `--add` file is appended after it. File names on the disk are derived from the source file stem, uppercased (e.g. `music.prg` → `MUSIC`).

After a successful build the compiler always prints a memory map:

```
demo.ub → demo.prg (386 bytes)

Load:    $080D - $0989

Variables (zero page):
  score   ZP:$08  byte
  lives   ZP:$0A  byte
  msg     ZP:$0C  string
  ptr     ZP:$0E  word

Subroutines:
  greet   $0900
  show    $0912

Arrays ($C000+):
  sprites $C000  8 bytes
```

With `-v` the output additionally shows the internal ZP allocations and a full hex dump.

Building

```
cargo build --release # binary: target/release/ub
cargo test           # unit + integration tests
```

Known limitations

Feature	Limitation
Integer arithmetic	8-bit unsigned (0–255); <code>word</code> vars hold 16-bit values
Subroutines	No recursion — ZP parameter slots are statically allocated
String vars	Read-only after init; assignment replaces the pointer, not the data
String concat runtime	<code>s1 + s2</code> prints sequentially — no heap allocation or length tracking
<code>rnd()</code> / <code>rnd(n)</code>	Simple LCG, not cryptographic; period = 256
<code>abs()</code> / <code>sgn()</code> / <code>min()</code> / <code>max()</code>	8-bit values only; <code>abs</code> / <code>sgn</code> treat values as signed (bit 7 = negative → <code>abs</code> two's-complements, <code>sgn</code> returns <code>\$FF</code>); <code>min</code> / <code>max</code> are unsigned (0–255)
<code>plot</code>	Out-of-range pixels are silently clipped ($Y \geq 200$ or $X \geq 320 \rightarrow$ no-op)
<code>mplot</code>	No bounds checking — x must be 0–159, y must be 0–199

Feature	Limitation
<code>plot4</code>	No bounds checking — x must be 0–79, y must be 0–49 (block mode)
<code>circle4</code>	Clips off-screen block pixels; useful radius is roughly 0–49 in 80×50 block mode
<code>chr\$</code>	No PETSCII↔ASCII mapping — n is passed as-is to CHROUT
<code>music play</code>	Requires <code>load sid</code> ; only one CIA1 wrapper is emitted (last <code>music play</code> wins)
Error reporting	Compile-time only; <code>onerr goto</code> handles KERNAL I/O errors at runtime